

Optimal Codes and Arcs for the Generalized Hamming Weights *

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Abstract

This text contains some notes on the generalized Griesmer bound, i.e. the Griesmer bound with respect to the generalized Hamming weight. We give a geometric characterization of minihypers with certain parameters. This implies that the associated codes are obtained by the classical Solomon–Stiffler construction. We present tables with the parameters of optimal binary codes for dimensions at most seven.

Key Words : linear codes, generalized Hamming weight, Galois geometry, additive codes, Griesmer bound.

1 Introduction

The Hamming weight of a codeword equals the size of its support and the minimum Hamming weight of a linear code is the minimum Hamming weight of the non-zero codewords. For a subcode the support is given by the set of all positions where at least one of the codewords in the subcode has a non-zero entry. The r th generalized Hamming weight of a linear code is the

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size of the smallest support of an r -dimensional subcode. The generalized Hamming weights can be used to describe the cryptography performance of a linear code over the wire-tap channel of type II [15] and to determine the trellis complexity of the code [3, 4]. From a geometrical point of view the r th generalized Hamming weight of a linear code corresponds to the minimum number of points outside of a subspace of codimension r , where the points are the one dimensional subspaces spanned by the columns of a generator matrix of the linear code. While one can easily find tables on the best known bounds for the parameters of linear codes with respect to the minimum Hamming distance for small parameters we were not able to find such a table for the r th generalized Hamming weight. The aim of this paper is to tabulate those numbers, where we will mainly use the geometric reformulation as multisets of points in projective spaces. In [1] the authors gave a construction for additive codes based on linear codes and their generalized Hamming weights. So, the constructions studied in this paper also give constructions for additive codes and the studied upper bounds show limitations for this specific construction.

The remaining part of the paper is structured as follows. In Section 2 we present the necessary preliminaries. In Section 3 we analyze the Griesmer bound for the r th generalized Hamming weight. In Section 4 we present two characterization results for minihypers with respect to the generalized Hamming weight that extend earlier results from [11]. In Section 5, we determine the minimum possible lengths of $[n, k]_q$ codes with given minimum r th generalized Hamming weight d for some small parameters. We present our results geometrically, i.e. we determine the maximum number $m_q^{(r)}(k, w)$ of points in $\text{PG}(k-1, q)$ such that each subspace of dimension r contains at most w points. We completely determine $m_2^{(r)}(k-1, w)$ for all $k \leq 7$, and all r .

2 Preliminaries

A $[n, k]_q$ code C is a k -dimensional subspace of $\mathbb{F}_q^n$. For $c = (c_1, \dots, c_n) \in \mathbb{F}_q^n$ we call $\text{supp}(c) := \{1 \leq i \leq n : c_i \neq 0\}$ the *support* of c and $\text{wt}(c) = |\text{supp}(c)|$ its *weight*. For a vector subspace $C \subseteq \mathbb{F}_q^n$, we define its *support* as the set of all coordinate positions in which the vectors of C are not identically zero. In other words, $\text{supp}(C) := \{1 \leq i \leq n : \exists c = (c_1, \dots, c_n) \in C, c_i \neq 0\}$. For two $\mathbb{F}_q$ -vector subspaces C, C' of $\mathbb{F}_q^n$ we write $C' \leq C$ if C' is a subspace of C . The r th generalized Hamming weight of a linear code C , denoted as $d_r(C)$, is the size of the smallest support of an r -dimensional subcode of C :

$$d_r(C) := \min\{|\text{supp}(C')| : C' \leq C, \dim(C') = r\}. \quad (1)$$

A linear code of length n , dimension k and r th generalized Hamming weight d_r will be called an $[n, k, d_r]^{(r)}$ -code. The sequence $(d_1(C), \dots, d_k(C))$ is

called the *weight hierarchy* of a linear $[n, k]_q$ code C . Clearly, $1 \leq d_1(C) < d_2 < \dots < d_k(C) \leq n$ (cf. [15]).

A *multiset* in $\text{PG}(k-1, q)$ is a mapping $\mathcal{K}: \mathcal{P} \rightarrow \mathbb{N}_0$, from the pointset $\mathcal{P}$ of $\text{PG}(r, q)$ to the set of non-negative integers, which assigns a multiplicity to each point of $\mathcal{P}$. This mapping is extended to any subset $\mathcal{Q}$ of $\mathcal{P}$ by $\mathcal{K}(\mathcal{Q}) = \sum_{P \in \mathcal{Q}} \mathcal{K}(P)$. The integer $\mathcal{K}(\mathcal{Q})$ is called the multiplicity of $\mathcal{Q}$.

A multiset $\mathcal{K}$ is called an (n, w) -*arc* (resp. (n, u) -*minihyper*), if $\mathcal{K}(\mathcal{P}) = n$, $\mathcal{K}(H) \leq w$ (resp. $\mathcal{K}(H) \geq u$) for each hyperplane H in $\text{PG}(k-1, q)$, and there is a hyperplane H_0 with $\mathcal{K}(H_0) = w$ (resp. $\mathcal{K}(H_0) = u$).

Let $\mathcal{K}$ be a multiset in $\text{PG}(k-1, q)$. We denote by w_r the maximal multiplicity of a r -dimensional subspace of $\text{PG}(r, q)$ with respect to $\mathcal{K}$:

$$w_r := \max \{ \mathcal{K}(S) : S \text{ is a subspace of } \text{PG}(k-1, q), \dim(S) = r \}. \quad (2)$$

Similarly, we denote by u_r the minimal multiplicity of a r -dimensional subspace of $\text{PG}(r, q)$ with respect to $\mathcal{K}$:

$$u_r := \min \{ \mathcal{K}(S) : S \text{ is a subspace of } \text{PG}(k-1, q), \dim(S) = r \}. \quad (3)$$

A multiset in $\text{PG}(k-1, q)$ of multiplicity n is called an $(n, w_r)^{(r)}$ -*arc* (resp. $(n, u_r)^{(r)}$ -*minihyper*) if each r -dimensional subspace of $\text{PG}(k-1, q)$ has multiplicity at most w_r (resp. at least u_r), and there exists an r -dimensional subspace with this multiplicity.

There exists a one-to-one correspondence between the isomorphism classes of the linear $[n, k]_q$ -codes of full length and the classes of projectively equivalent multisets in $\text{PG}(k-1, q)$, where $k \geq 2$. Clearly,

$$w_r + d_{k-r-1} = n. \quad (4)$$

This implies $w_0 < w_1 < \dots < w_{k-1} = n$. If $s \geq w_0$ is an integer, then $\mathcal{K}' = s - \mathcal{K}$ is a multiset of cardinality $sv_k - n$ with $u_r = sv_{r+1} - w_r$, $r = 0, 1, \dots, k-1$. Here $v_r = (q^r - 1)/(q - 1)$, as usual.

Definition 1. We denote by $m_q^{(r)}(k, w)$ the maximal multiplicity of a multiset in $\text{PG}(k-1, q)$ such that every r -dimensional subspace has multiplicity at most w . In other words, $m_q^{(r)}(k, w)$ is defined as the largest n for which there exists an $(n, w)^{(r)}$ -arc in $\text{PG}(k-1, q)$.

3 The Griesmer bound

A result analogous to the classical Griesmer bound using the generalized Hamming weights was proved in [6, 8]. The corresponding bound is called *the generalized Griesmer bound*.

Theorem 1. *Let C be an $[n, k, d_r]^{(r)}$ -code, where $1 \leq r \leq k$. Then*

$$n \geq d_r + \sum_{i=1}^{k-r} \left\lceil \frac{d_r}{q^i v_r} \right\rceil. \quad (5)$$

We shall denote the left-hand side of (5) by $g_q^{(r)}(k, d_r)$. In case of $r = 1$, we write simply $g_q(k, d)$. The following result is well-known (see e.g. [12]).

Theorem 2. *Let $\mathcal{K}$ be an $(n, n - d)$ -arc in $\text{PG}(k - 1, q)$ and let C be an $[n, k, d]_q$ code associated with $\mathcal{K}$. If $n = t + g_q(k, d)$, where $t \geq 0$ is an integer, then*

$$w_j \leq t + \sum_{i=k-1-j}^{k-1} \left\lceil \frac{d}{q^i} \right\rceil \quad (6)$$

If d is written as

$$d = \sigma q^{k-1} - \sum_{i=0}^{k-2} \varepsilon_i q^i \quad (7)$$

then

$$g_q^{(1)}(k, q) = \sigma v_k - \sum_{i=0}^{k-2} \varepsilon_i v_{i+1}. \quad (8)$$

Let C be an $[n = g_q(k, d), k, d]_q^{(1)}$ -code and let $\mathcal{K}$ an arc $(n, n - d)$ -arc in $\text{PG}(k - 1, q)$ associated with C . By Theorem 2,

$$w_{k-1-j} = \sigma v_{k-j} - \sum_{i=j}^{k-2} \varepsilon_i v_{i+1-j}. \quad (9)$$

Since the maximal multiplicity of a point with respect to $\mathcal{K}$ is σ , $\mathcal{K}$ can be represented as $\mathcal{K} = \sigma \chi_{\mathcal{P}} - \mathcal{F}$, where $\mathcal{F}$ is a minihyper with parameters

$$\left(\sum_{i=0}^{k-2} \varepsilon_i v_{i+1}, \sum_{i=0}^{k-2} \varepsilon_i v_i \right) \quad (10)$$

with maximal multiplicity of a point at most σ . Such minihypers without the restriction on the maximal point multiplicity can always be constructed as a sum of subspaces: ε_{k-2} hyperplanes, ε_{k-3} hyperlines, and so on, ε_1 lines and ε_0 points. Such minihypers are called *canonical*.

A canonical minihyper $\mathcal{F}$ in $\text{PG}(k-1, q)$ is said to be of type $\sum_{i=1}^{k-2} \varepsilon_i [i-1]$ if it is the sum ε_{k-2} hyperplanes, ε_{k-3} -hyperlines, and so on ε_2 lines, and ε_1 points. In other words, $\mathcal{F} = \sum_i \chi_{S_i}$, where exactly ε_j of the subspaces S_i are of (projective) dimension $j - 1$. The arc $\mathcal{K}$ obtained by subtracting $\mathcal{F}$ from s copies of $\text{PG}(K - 1, q)$ will be said to have type $s[k - 1] - \sum_{i=1}^{k-2} [\varepsilon_i]$.

Minihypers with parameters

$$\left(\sum_{i=r}^{k-2} \varepsilon_i v_{i+1}, \sum_{i=r}^{k-2} \varepsilon_i v_{i+1-r} \right)^{(k-1-r)} \quad (11)$$

can be constructed in a similar fashion as canonical minihypers.

Theorem 3. *Let $\mathcal{F}$ be a minihyper in $\text{PG}(k-1, q)$ with parameters given by (11), $0 \leq \varepsilon_i \leq q-1$, and with maximal point multiplicity w_0 . Then for every $\sigma \geq w_0$ $\mathcal{K} = \sigma - \mathcal{F}$ is a (multi)arc in $\text{PG}(k-1, q)$ with parameters*

$$\left(n = \sigma v_k - \sum_{i=r}^{k-2} \varepsilon_i v_{i+1}, w_{k-r-1} = \sigma v_{k-r} - \sum_{i=r}^{k-2} \varepsilon_i v_{i+1-r} \right)^{(k-1-r)}.$$

The code associated with $\mathcal{K}$ is a Griesmer code with $n = g_q^{(r)}(k, d_r)$.

If the maximal point multiplicity of $\mathcal{F}$ with parameters given by (11) is large, such minihypers can always be constructed. Thus we have the following theorem, essentially due to Solomon and Stiffler [13].

Theorem 4. *If $d_r \geq (q-1)q^{k-1}$ Griesmer $[n = g_q^{(r)}(k, d_r), k, d_r]_q^{(r)}$ -codes do exist for all q, k and r .*

If a code attains the classical Griesmer bound, it attains also the Griesmer bound for all generalized weights.

Theorem 5. *Let C be an $[n = g_q(k, d), k, d]_q^{(1)}$ -code where $d^{(1)} = d$ is given by (7). Then*

$$d_r(C) = v_r \cdot \left(\sigma q^{k-r} - \sum_{i=r}^{k-1} \varepsilon_{i-1} q^{i-r} \right) - \sum_{i=1}^{r-1} \varepsilon_{i-1} v_i,$$

and C attains the Griesmer bound for the r th generalized Hamming weight.

4 A Characterization Result

It was proved in [11] that every (xv_{k-1}, xv_{k-2}) -minihyper in $\text{PG}(k-1, q)$, $q = p^h$ is canonical for all $x \leq q - q/p$. This result remains true also for the generalized Hamming weights.

Theorem 6. *Every $(xv_{k-1}, xv_r)^{(r)}$ -minihyper in $\text{PG}(k-1, q)$, $q = p^h$, $k \geq 4$, with $x \leq q - \frac{q}{p}$ is a sum of hyperplanes for all $r \in \{1, \dots, k-2\}$.*

Proof. Let us note that the case $r = k - 2$ was proved in [11]. Below we shall prove the result for $r = k - 3$. The remaining values of r are treated in a similar fashion.

Let $\mathcal{F}$ be a $(xv_{k-1}, xv_{k-3})^{(k-3)}$ -minihyper. Since every $(k-3)$ -space is of multiplicity xv_{k-3} , it follows that the multiplicity of every hyperplane is at least xv_{k-2} . If H is a minimal hyperplane then $\mathcal{F}|_H$ is a $(xv_{k-2}, xv_{k-3})^{(k-3)}$. Hence it is a sum of hyperlines.

Fix a minimal hyperplane H_0 and consider a minimal $(k-3)$ -subspace S in H_0 . Denote by H_i , $i = 0, \dots, q$, the hyperplanes through S . A counting argument gives that $\mathcal{F}(H_i) = xv_{k-2}$ for all i . The restriction of $\mathcal{F}$ on any of the hyperplanes H_i is a sum of x $(k-3)$ -dimensional subspaces.

Consider a projection φ from a $(k-4)$ -dimensional subspace $\delta \subset S$ without 0-points onto some plane π disjoint from δ . Since $\mathcal{F}|_{H_i}$ is a sum of $(k-3)$ -dimensional subspaces, $\mathcal{F}(\delta) = tv_{k-3} + (x-t)v_{k-4}$. Set $L_i = \varphi(H_i)$. This line is of type

$$(t_1q^{k-3} + (x-t)q^{k-4}, \dots, t_qq^{k-3} + (x-t)q^{k-4}, (x-t)q^{k-4})$$

with respect to $\mathcal{F}^{(\varphi)}$, where $\sum t_i = t$.

Define a new multiset

$$\mathcal{G} = \frac{\mathcal{F}^{(\varphi)} - (x-t)q^{k-4}}{q^{k-3}}.$$

The line L_i is of type $(t_1, \dots, t_q, 0)$. Moreover, the multiset $\mathcal{G}$ is a $(t(q+1), t)$ -blocking set, which is a sum of lines. Going back to $\mathcal{F}$, we deduce that there exists a hyperplane G through δ such that $\mathcal{F} = \mathcal{F}' + \chi_G$ where $\mathcal{F}'$ is a $((x-1)v_{k-1}, (x-1)v_r)^{(r)}$ -minihyper, and χ_G is the characteristic function of G . Repeating this argument we obtain

$$\mathcal{F} = \chi_{G_1} + \dots + \chi_{G_x},$$

where $G_1, \dots, G_x$ are (not necessarily) different hyperplanes. $\square$

Similar arguments can be used to prove the following theorem.

Theorem 7. *Every $(xv_{k-1} + v_{k-2}, xv_r + v_{r-1})^{(r)}$ in $\text{PG}(k-1, p)$, p prime, for $x \leq p-2$ is the sum of x hyperplanes and one subspace of codimension 2.*

The problem of characterizing the minihypers with parameters (11) is very difficult. The fact that all minihypers with certain parameters (11) are canonical means that all codes associated with this minihyper are obtained by the Solomon-Stiffler construction [13].

5 Exact values

In this section, we tackle the problem of finding the value of $m_q^{(r)}(k-1, w)$ for fixed k, r, w and q . Some instances of this problem are of special interest. So, the case $r = 1$ is the problem of determining the largest size of a generalized cap, – a multiset of points such that each line has multiplicity at most w (cf. [2]). If $w = 2$ this is the notorious maximal cap problem.

From the connection between the linear $[n, k]_q$ codes and the multisets of points in $\text{PG}(k-1, q)$, we get that $m_q^{(r)}(k-1, w)$ is the largest integer n such that $n \geq g_q^{(k-r-1)}(k, n-w)$. This integer will be called the *Griesmer upper bound* for $m_q^{(r)}(k-1, w)$.

Another bound on $m_q^{(r)}(k-1)$ is the following. Set $s_r = w$ and $s_{r+i} = m_q^{(r+i-1)}(r+i, s_r)$, $i = 0, \dots, k-1-r$. Then $m_q^{(r)}(k-1) \leq s_{k-1}$. We call this bound the *coding upper bound* for $m_q^{(r)}(k-1, w)$.

Some of our results for the binary case for dimensions $k = 6, 7$ are presented in the tables below. For all tables with $k \leq 7$, see [10].

w	$m_2^{(2)}(6, w)$	construction	upper bound
3	11	[10]	[10]
4	64	[6] – [5]	Griesmer upper bound
5	65	plus point	Griesmer upper bound
6	96	[6] – [4]	Griesmer upper bound
7	127	[6]	Griesmer upper bound
8	128	plus point	Griesmer upper bound
9	129	plus point	Griesmer upper bound
10	160	2[6] – [5] – [4]	Griesmer upper bound

Table 1: Exact values for $m_2^{(2)}(6, w)$.

w	$m_2^{(2)}(5, w)$	construction	upper bound
3	8	[10]	[10]
4	32	[5] – [4]	Griesmer upper bound
5	33	plus point	Griesmer upper bound
6	48	[5] – [3]	Griesmer upper bound
7	63	[5]	Griesmer upper bound
8	64	plus point	Griesmer upper bound
9	65	plus point	Griesmer upper bound
10	80	2[5] – [4] – [3]	Griesmer upper bound

Table 2: Exact values for $m_2^{(2)}(5, w)$.

w	$m_2^{(3)}(5, w)$	construction	upper bound
4	7	projective base	[10]
5	11	BKLC/ILP	Coding upper bound
6	18	BKLC/ILP	Griesmer upper bound
7	19	plus point	Coding upper bound
8	32	[5] – [4]	Griesmer upper bound
9	33	plus point	Griesmer upper bound
10	34	plus point	Griesmer upper bound
11	38	BKLC/ILP	[10]
12	48	[5] – [3]	Griesmer upper bound
13	49	plus point	Griesmer upper bound
14	56	[5] – [2]	Griesmer upper bound
15	63	[5]	Griesmer upper bound
16	64	plus point	Griesmer upper bound
17	65	plus point	Griesmer upper bound
18	66	plus point	Griesmer upper bound
19	73	BKLC/ILP	Griesmer upper bound
20	80	2[5] – [4] – [3]	Griesmer upper bound
21	81	plus point	Griesmer upper bound
22	88	2[5] – [4] – [2]	Griesmer upper bound
23	95	sum construction	Griesmer upper bound
24	96	plus point	Griesmer upper bound
25	97	plus point	Griesmer upper bound
26	104	2[5] – [3] – [2]	Griesmer upper bound
27	111	sum construction	Griesmer upper bound
28	112	plus point	Griesmer upper bound
29	119	sum construction	Griesmer upper bound
30	126	sum construction	Griesmer upper bound
31	127	plus point	Griesmer upper bound
32	128	plus point	Griesmer upper bound
33	129	plus point	Griesmer upper bound
34	136	3[5] – [4] – [3] – [2]	Griesmer upper bound

Table 3: Exact values for $m_2^{(3)}(5, w)$.

w	$m_2^{(3)}(6, w)$	construction	upper bound
4	9	ILP	[10]
5	19	BKLC/ILP	[10]
6	28	ILP	[10]
7	35	BKLC/ILP	Coding upper bound
8	64	[6] – [5]	Griesmer upper bound
9	65	plus point	Griesmer upper bound
10	66	plus point	Griesmer upper bound
11	72	[10]	[10]
12	96	[6] – [4]	Griesmer upper bound
13	97	plus point	Griesmer upper bound
14	112	[6] – [3]	Griesmer upper bound
15	127	[6]	Griesmer upper bound
16	128	plus point	Griesmer upper bound
17	129	plus point	Griesmer upper bound
18	130	plus point	Griesmer upper bound
19	145	BKLC/ILP	Griesmer upper bound
20	160	2[6] – [5] – [4]	Griesmer upper bound
21	161	plus point	Griesmer upper bound
22	176	2[6] – [5] – [3]	Griesmer upper bound
23	191	sum construction	Griesmer upper bound
24	192	sum construction	Griesmer upper bound
25	193	sum construction	Griesmer upper bound
26	208	2[6] – [4] – [3]	Griesmer upper bound

Table 4: Exact values for $m_2^{(3)}(6, w)$.

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